4B Rates, Speed and Best Buys

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Express each relationship as a simplified unit rate:

(a) $360\ \text{km}$ in $4.5\ \text{hours}$ (km/h)

(b) $\$138$ for $6\ \text{hours}$ of work ($\$/\text{h}$)

(c) $42\ \text{L}$ of petrol for $525\ \text{km}$ (km/L)

(d) $750\ \text{words}$ typed in $12\ \text{minutes}$ (words/min)

Q2. Calculate the distance, average speed, or time required:

(a) Distance travelled at $96\ \text{km/h}$ for $2.5\ \text{hours}$

(b) Time taken to travel $315\ \text{km}$ at $90\ \text{km/h}$

(c) Average speed for $200\ \text{m}$ in $25\ \text{seconds}$ (m/s)

(d) Distance run at $6.5\ \text{m/s}$ in $40\ \text{seconds}$

Q3. Convert each speed or rate to the specified unit:

(a) $15\ \text{m/s}$ to km/h

(b) $108\ \text{km/h}$ to m/s

(c) $25\ \text{m/s}$ to km/h

(d) $30\ \text{L/min}$ to L/s

Q4. Calculate the unit price per $100\ \text{g}$ or per $1\ \text{L}$ for each grocery item:

(a) $250\ \text{g}$ of cheese for $\$4.50$ ($\text{per } 100\ \text{g}$)

(b) $400\ \text{g}$ of oats for $\$2.60$ ($\text{per } 100\ \text{g}$)

(c) $1.5\ \text{L}$ of olive oil for $\$18.90$ ($\text{per } 1\ \text{L}$)

(d) $600\ \text{g}$ of honey for $\$9.30$ ($\text{per } 100\ \text{g}$)

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A supermarket sells Australian almonds in three pack sizes: Pack A ($200\ \text{g}$ for $\$5.20$), Pack B ($500\ \text{g}$ for $\$11.50$), and Pack C ($800\ \text{g}$ for $\$19.20$). Calculate the cost per $100\ \text{g}$ for each pack and determine the best buy.

Q6. A family car consumes $6.8\ \text{L}$ of petrol per $100\ \text{km}$ on the Hume Freeway. (a) How many litres of petrol are needed for a $450\ \text{km}$ drive? (b) At $\$1.95\ \text{per litre}$, what is the fuel cost for the trip?

Q7. A V/Line train travels $75\ \text{km}$ from Melbourne to Geelong in $50\ \text{minutes}$ ($rac56\ \text{hour}$). Calculate the train's average speed in km/h and in m/s.

Q8. Tap A fills a $600\ \text{L}$ trough at $12\ \text{L/min}$, and Tap B fills the same trough at $8\ \text{L/min}$. If both taps are turned on together, how many minutes does it take to fill the $600\ \text{L}$ trough?

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. A cyclist rides $30\ \text{km}$ uphill at $15\ \text{km/h}$ ($2\ \text{hours}$) and returns $30\ \text{km}$ downhill along the same road at $30\ \text{km/h}$ ($1\ \text{hour}$). Show that her overall average speed for the $60\ \text{km}$ round trip is $20\ \text{km/h}$, NOT $\dfrac{15+30}{2} = 22.5\ \text{km/h}$.

Q10. Explain why converting $1\ \text{m/s}$ to km/h multiplies by 3.6: show where the 3600 and 1000 come from in the unit conversion.